

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Bachelor of Pharmacy (B. Pharm)

Semester-V University Backlog Examination April- 2016

PH 303 Medicinal Chemistry-I

Date: 18/04/2016, Monday,

Time: 10:00 a.m to 01:00 p.m

Maximum Marks: 80

Instructions:

1. Figures to right indicate marks.
2. Section wise separate answer book is to be used.
3. Draw neat sketches wherever necessary.

SECTION- I

- Q 1** Attempt any TWO of the following:
- A** Explain a detailed note on Phase-I metabolic reactions. **10**
- B** Answer the Following:
- 1) Add in detail importance of Cytochrome in Drug Metabolism **05**
- 2) Discuss the relationship between biological action and physicochemical properties for Hydrogen Bonding with suitable Examples **05**
- C** 1) Write a short note on Pro Drugs **05**
- 2) Give the structure of: (ANY FIVE) **05**
(1) Quinoline (2) Tetrazole (3) Furan (4) Imidazole (5) Oxazole (6) Pteridine
- Q 2** Attempt any TWO of the following:
- A** 1) Write a Comments on: (ANY THREE) **06**
- I. Pyridine is more stronger base than Pyrrole.
- II. Electrophilic substitution takes place at 2nd and 5th position in Pyrrole.
- III. Furan is less aromatic than Thiophene.
- IV. Impurities of Thiophene in benzene can be removed by cold sulphuric acid.
- V. Proton pump inhibitors act as pro-drug.
- 2) Give Proton Transfer (Hantzsch) synthesis for Pyridine with reaction mechanism. **04**
- B** Answer the Following:
- 1) Give methods of preparation of any TWO of the following: **05**
1) Pyrrole (2) Thiophene (3) Imidazole
- 2) Explain Fischer's Indole synthesis with reaction mechanism. **05**

- C Answer the Following:**
- 1) Write Skraup synthesis with reaction mechanism. 05
 - 2) Write Paal-Knorr synthesis for Pyrrole with reaction mechanism. 05

SECTION- II

Q 3 Attempt any TWO of the following:

- A Answer the Following:**
- 1) What are Parasympathomimetic Agents? Classify them with examples. 06
 - 2) Give structure, chemical name and synthesis of following compounds. 04
 - (i) Nephazoline
 - (ii) Pentoprazole

- B Answer the Following:**
- 1) Explain Irritable Bowel Syndrome. Give the details of drugs used in the treatment of it. 05
 - 2) Give SAR of Beta-blockers. 05

- C Answer the Following:**
- 1) What is Asthma? Explain Anti-asthmatic drugs in detail. 05
 - 2) Give SAR of H₂-Antagonists. 05

Q 4 Attempt any TWO of the following:

- A Answer the Following:**
- 1) Explain the following terms with example. 06
 - (i) Laxatives
 - (ii) Prokinetics
 - (iii) Antacid
 - 2) Give SAR of Muscarinic Antagonists. 04

- B Answer the Following:**
- 1) What are proton pumps Inhibitors? Explain its mechanism of action. Give Synthesis of Omeprazole. 06
 - 2) Give two suitable Examples and Structure of : 04
 - (i) H₁-Antagonists
 - (ii) Neuromuscular Blocking Agents

- C Answer the Following:**
- 1) What are Sympathomimetic Agents? Classify them with examples. 05
 - 2) Write synthesis of Salbutamol and Atenolol. 05